Recursive quantitative eventual-period bounds in Divisor Nim

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Editorial note. Complete original mathematical paper, all source-local proof machinery and embedded bibliography retained unchanged. Authors retain responsibility for mathematics; no independent author review, proof repair, added hypotheses, counterexample search or certification.

Editorial note. The heap-count-independent SG ceiling and quantitative recursive eventual-period bound are distinct targets. Closed-form/refined/prime-period and joined-board consequences remain complete original support/context, not extra counts.

Editorial note. Original finite-computation reports and final conjectures retained verbatim without agent rerun, certification or counting conjectures as proved. Smallest fixed heap, largest fixed heap and the varying heap remain distinct as in the author source.

Editorial note. Rendering assembly correction only: prior broad proof spans overlapped the separately selected statement and trailing context, duplicating source labels in the rendered document. Proof/context ranges are now partitioned at the exact statement boundary, so every original line89–1868 appears exactly once. No author label, mathematical text or original source changed; failed r001 retained.

Allowed standard prerequisites (editorial).

- Divisor Nim legal moves, Sprague-Grundy mex recursion,2-adic depth and eventual periods. Original shared divisor rule across all other nonempty heaps, single-heap convention, smallest versus largest fixed heap and variable-heap exclusion conventions retained. Recurrence U_k, range C, spacing W/J and recursive successor periods are fully defined in retained context.. Complete original analytic bounds and all their supporting proofs supplied; author finite computational reports and unproved conjectures remain clearly original context, without agent computation or certification.

1 Can one small heap control a large board?

A board contains a heap of size 1. We add many more heaps, each as large as we wish. Can the Grundy value grow without limit? In Divisor Nim, it cannot. While the heap 1 remains unchanged, every removal from another heap must divide 1 and therefore must equal 1. We will show that the value stays at most 2, then extend the argument to any smallest heap.

The same idea leads to three questions. How large can the Grundy value be? How long can a period of these values be when one heap varies? What information survives when boards are joined? We address them in that order.

The rules and the value being bounded

A position $P = (h_1, \ldots, h_m)$ lists the positive heap sizes ($h_i \geq 1$). A move subtracts a positive integer d from one heap h_i, subject to

$$d \leq h_i, \qquad d \mid h_j \quad \text{for every } j \neq i. \tag{1}$$

Here $d \mid h_j$ means that d divides h_j. The removal need not divide the heap being changed. Empty heaps impose no restriction, since every positive integer divides 0; we omit them from the displayed board. Thus m denotes the current number of nonempty heaps. If only one heap remains then every removal up to its

size is permitted. Players alternate, and the player with no legal move loses. These are Morisawa's rules for m-pile Divisor Nim [1].

A *successor* is a board left after one legal move. The *Sprague–Grundy (SG) value* of a board is defined recursively [3]. The empty board has value 0. For another board, list the values of its successors and choose the smallest nonnegative integer missing from the list:

$$g(P) = \operatorname{mex}\{g(Q) : P \longrightarrow Q\}. \tag{2}$$

The notation mex means the smallest missing nonnegative integer. A board is losing exactly when its SG value is 0, and winning when its SG value is positive. A sole heap of size h has SG value h.

The distinction between moves and values is essential here. A thousand moves to boards with SG value 0 still supply only one value to the mex.

Even a very large successor SG value need not make the current board's SG value large. For example, the only successors of $(1, 100)$ are (100) and $(1, 99)$, with SG values 100 and 0. Therefore

$$g(1, 100) = \operatorname{mex}\{g(100), g(1, 99)\} = \operatorname{mex}\{100, 0\} = 1.$$

Morisawa's bounds and periodicity

The SG bound. Morisawa counts possible removals from a reference heap h_i and from the other $m - 1$ heaps. Each move supplies at most one successor SG value, so the total number of moves bounds the mex.

Quantity and meaning	Expression and reason
h_i Size of the reference heap	A removal must fit in the heap: $1 \le d \le h_i$. There are therefore at most h_i possible removals.
$\tau(h_i)$ Number of positive divisors of h_i	Every removal from another heap must divide h_i. There are therefore at most $\tau(h_i)$ possible removal amounts from each other heap.
$(m-1)\tau(h_i)$ Bound for all other heaps	There are $m - 1$ other heaps, each permitting at most $\tau(h_i)$ removals. Multiplying gives their combined bound.

Adding the contributions and choosing the reference heap that gives the smallest bound yields Morisawa's general SG bound [1, Lemma 5.2]:

$$\boxed{g(P) \le \min_i \big[h_i + (m-1)\tau(h_i)\big].} \tag{3}$$

For a simpler expression, use the smallest heap, $h_{\min}$. Since $\tau(h_{\min}) \le h_{\min}$,

$$g(P) \le h_{\min} + (m-1)h_{\min} = mh_{\min}.$$

The period bound. The board (A, n) consists of fixed heaps A and one varying heap of size n. An *eventual period* ω_A is a positive increase in the varying heap size that preserves the SG value from some starting size n onward:

$$\boxed{g(A, n+t) = g(A, n+\omega_A+t) \qquad \text{for every integer } t \ge 0.}$$

The smallest such positive increase is the *least eventual period*. We use ω_A for this least period when stating divisibility claims; any positive multiple of it is also an eventual period. Morisawa proves that an eventual period exists whenever at least one fixed heap is nonempty [1, Theorem 2.5].

For the construction below, $A = (h_1, \ldots, h_{m-1})$ contains at least two nonempty heaps. The number m includes the varying heap, and $h_{\min}$ is the smallest fixed heap.

Morisawa's period bound combines the number of possible blocks of consecutive SG values with the comparison spacing ω'.

A move on the varying heap reaches $(A, n-d)$, using an earlier SG value supplied by the block. A move on a fixed heap produces a smaller fixed board Q and reaches (Q, n). The comparison spacing makes legal moves and these smaller-board SG values repeat together.

Quantity and meaning	Expression and reason
$0, \ldots, mh_{\min}$ SG value range	The SG bound gives $g(A,n) \le mh_{\min}$. Including zero gives $mh_{\min}+1$ possibilities for each entry in a block.
M_A Increase in n that repeats legal moves	Omit each fixed heap in turn, take the GCD of the remaining fixed heaps, then take their LCM: $$M_A = \operatorname{lcm}_{1\le i\le m-1} \gcd\{h_j : 1 \le j \le m-1,\ j \ne i\}.$$ Every possible removal size divides M_A. Thus $d \mid n$ exactly when $d \mid n + M_A$, and increasing a sufficiently large n by M_A preserves legal moves.
$M_A + 1$ Number of consecutive SG values compared	Morisawa compares blocks $$\big(g(A,n), \ldots, g(A, n+M_A)\big).$$ Every removal from the varying heap is at most M_A, so these entries cover all earlier values needed to continue the comparison.
$\omega_{Q_1}, \ldots, \omega_{Q_t}$ SG periods for all smaller fixed boards	The list $Q_1, \ldots, Q_t$ contains every nonempty fixed board with fewer total stones than A and at most $m-1$ heaps. Morisawa first establishes an eventual period ω_{Q_i} for each sequence $g(Q_i, n)$. This covers the values reached by moves on fixed heaps.
ω' Comparison spacing	The required common multiple is $$\omega' = \operatorname{lcm}\big(M_A, \omega_{Q_1}, \ldots, \omega_{Q_t}\big).$$ It repeats legal moves and all the smaller-board SG values together, once their repeating parts have begun.

There are at most $(mh_{\min}+1)^{M_A+1}$ different blocks. Comparing one more block than this number, at intervals of ω', guarantees two matching blocks. Both types of moves then supply matching SG values, so the agreement continues indefinitely. Their separation is an eventual period, giving Morisawa's counting bound [1, Section 8]:

$$\boxed{\omega_A \le \underbrace{\omega'}_{\text{comparison spacing}} \underbrace{(mh_{\min}+1)^{M_A+1}}_{\text{number of possible blocks}}.} \tag{4}$$

Our contribution. We obtain an SG bound that depends only on the smallest heap, with no dependence on the number or sizes of the remaining heaps.

For periodicity, the smallest fixed heap bounds the SG value range, the number of consecutive SG values compared, and an increase in n that repeats legal moves. The remaining dependence on the other heaps is confined to the SG periods of direct successor boards.

We then replace these unknown periods with a common-period construction. The largest fixed heap keeps the same bounds valid throughout the successor chains. Each recursive layer can add a bounded multiplier; the total fixed stones bound how many layers are needed. This gives an explicit period and a fixed finite set of possible prime divisors. Their exponents may still grow with the number of layers. The construction also gives a period bound for joined boards.

The title refers to removing the heap-count parameter m from Morisawa's bounds. We achieve this for the SG value. For the period, we remove m from the local terms and identify the remaining dependence through recursive layers. Our conjecture proposes the common period $2\operatorname{lcm}(1, \ldots, h_{\max})$, depending only

on the largest fixed heap. Morisawa's two papers provide the starting point [2, 1]; the odd-heap cases lead to our bounds for all-even boards with arbitrarily many heaps.

2 Bounding Grundy values

The SG value is bounded by the number of distinct successor values, rather than the number of moves. Many moves reaching the same value contribute only once.

We start with positions containing odd heaps. For all-even positions, we then classify removals by their factors of 2. Some moves reach positions already bounded, others reach value 0, and the remaining moves are few enough to count. Adding these contributions gives the SG bound.

The starting cases: odd heaps

An unchanged odd heap forbids every even removal. Therefore, when at least two heaps are odd, every legal removal is odd and changes the number of odd heaps by one.

Lemma 2.1 (Odd heaps)**.** *Among positions containing an odd heap, those with an even number of odd heaps are losing. Those with an odd number are winning. When at least two heaps are odd, the exact SG values are:*

Number of odd heaps	*SG value*
Even, and at least two	0
Odd, and at least three	1

Proof. The outcome statement is the odd-heap case of Morisawa's criterion [1, Theorem 2.2]. From an odd number of at least three odd heaps, every move leaves a positive even number of odd heaps. Every successor therefore has value 0, so the current value is $\operatorname{mex}\{0\} = 1$. □

Exactly one odd heap requires a bound rather than this exact formula. Moves on any other heap create a second odd heap and reach value 0. Only moves on the odd heap can contribute additional values.

The symbol H denotes a fixed upper limit on the size of one positive heap. Keeping H fixed lets the same bound apply after that heap shrinks. Every legal removal then satisfies

$$d \le H. \tag{5}$$

Indeed, a removal from that small heap must fit inside it. A removal from another heap must divide the unchanged small heap, so it also cannot exceed its size.

Lemma 2.2 (The starting bound)**.** *If a position contains an odd heap and a positive heap of size at most H, then*

$$g(P) \le H + 1.$$

Proof. If at least two heaps are odd, Lemma 2.1 gives $g(P) \le 1$.

If exactly one heap is odd and other heaps remain, all moves on those other heaps reach value 0. There are at most H removal sizes available on the odd heap, by (5). Thus there are at most $H + 1$ distinct successor values, giving $g(P) \le H + 1$.

If only one heap remains, its size is at most H, and its SG value equals its size. □

Classifying moves by 2-adic depth

The *2-adic depth* $v_2(a)$ of a positive integer a is the number of factors of 2 in it:

$$a = 2^{v_2(a)}u, \qquad u \text{ odd}.$$

Equivalently, this counts how many times a can be halved before becoming odd.

For a nonempty position, its minimum depth is $k(P)$, and the number of heaps at that depth is $r(P)$:

$$k(P) = \min_i v_2(h_i), \qquad r(P) = \#\{i : v_2(h_i) = k(P)\}. \tag{6}$$

Dividing every heap by $2^{k(P)}$ exposes exactly $r(P)$ odd heaps.

The smallest heap need not have minimum depth. For example, in $(4, 6)$ the smallest heap is 4, but 6 has the smaller 2-adic depth.

Proposition 2.3 (Morisawa's outcome criterion)**.** *A nonempty position is losing exactly when the number of heaps at minimum depth is even:*

$$\boxed{g(P) = 0 \iff r(P) \text{ is even.}} \tag{7}$$

Consequently, dividing all heaps of an all-even position by 2 preserves its winning or losing outcome.

Proof. The criterion is Morisawa's Theorem 2.2 [1]. Common halving decreases every heap's depth by one and leaves the number attaining the minimum unchanged. □

Common halving need not preserve the full SG value. For example,

$$g(2, 4) = 3, \qquad g(1, 2) = 1.$$

To bound the full value, we account for the different removals.

Consider an all-even winning position. Its minimum depth $k = k(P)$ is positive, and $r(P)$ is odd. A removal of size d has depth $j = v_2(d)$. Its contribution is determined by how j compares with k and which heap is changed.

Condition and meaning	Successor and reason
$j < k$ Removal below the minimum depth	The changed heap satisfies $v_2(h_i - d) = j$. After division by 2^j, an even integer loses an odd integer, leaving an odd remainder. The successor has lower minimum depth, where we use the earlier bound.
$j = k$ A minimum-depth heap remains unchanged	Division by 2^k makes the removal odd, so the count at depth k changes by one. An unchanged heap keeps that count positive. It is now even, so the successor has SG value 0.
$r(P) = 1, \quad 2^k \mid d$ Removal from the unique minimum-depth heap	The preceding two arguments do not bound this successor's value. We instead count the moves: each removal amount contributes at most one additional SG value.

Table 1: The three types of moves from an all-even winning position.

A removal with $j > k$ cannot divide an unchanged heap at depth k. It must therefore change the unique heap at that depth, so it belongs to the third row. The three rows cover every move.

Counting the contributions using one small heap

The fixed size limit H controls both parts of the argument. It bounds the removal sizes, and a positive heap of size at most H survives every move to which we apply the lower-depth bound.

To see this, suppose the current position contains a heap $a \le H$ and a removal has depth $j < k$. If another heap is changed, a remains. If a is changed, deleting it would require $d = a$, which is impossible because

$$v_2(d) = j < k \le v_2(a).$$

A positive remainder smaller than a therefore survives.

Now consider the third row of Table 1. These moves all act on one particular heap: the unique heap of minimum depth. Their removal sizes are multiples of 2^k and cannot exceed H. Hence

$$\text{number of these moves} \le \left\lfloor \frac{H}{2^k} \right\rfloor. \tag{8}$$

Each size gives at most one move because the target heap is fixed. This is why the count has no factor for the number of heaps.

These moves may delete the small heap or reach a board with a large SG value. We do not calculate their successor values: each move contributes at most one value.

Write $U_j(H)$ for the bound we construct at minimum depth j, for positions containing a positive heap of size at most H. The odd-heap result supplies the starting value

$$U_0(H) = H + 1.$$

At depth $k \geq 1$, we add two counts: the possible values already covered at lower depths, including 0, and the additional values from removals on the unique minimum-depth heap.

Quantity and meaning	Expression and reason
$U_{k-1}(H) + 1$ Number of values in the lower-depth range	The lower-depth successors and losing successors have values in $0, \ldots, U_{k-1}(H)$. Including zero gives at most $U_{k-1}(H) + 1$ distinct values.
$\lfloor H/2^k \rfloor$ Number of multiples of 2^k up to H	These removals are multiples of 2^k no larger than H. There are at most $\lfloor H/2^k \rfloor$ such amounts. Each acts on the same heap and contributes at most one successor value.

Table 2: The two contributions to the bound at depth k.

The mex is at most the number of distinct successor values. Adding the two contributions gives

$$U_0(H) = H + 1, \qquad \boxed{U_k(H) = \underbrace{U_{k-1}(H) + 1}_{\text{lower-depth range, including 0}} + \underbrace{\left\lfloor \frac{H}{2^k} \right\rfloor}_{\text{additional values}}}. \tag{9}$$

The recursion follows decreasing 2-adic depth, not a sequence of moves in actual play. We apply it only to the first row of Table 1; the other rows are accounted for without further recursion.

Theorem 2.4 (A bound independent of heap count)**.** *Every nonempty position containing a heap of size at most H satisfies*

$$g(P) \leq U_{k(P)}(H).$$

If the number $r(P)$ of heaps at minimum depth is odd and at least three, then

$$g(P) \leq U_{k(P)}(H) - \left\lfloor \frac{H}{2^{k(P)}} \right\rfloor. \tag{10}$$

Proof. A sole heap has value equal to its size, at most H, so it satisfies the bound at every depth.

For positions with more heaps, induct on the minimum depth. At depth zero, Lemma 2.2 gives $g(P) \leq H + 1 = U_0(H)$.

At depth $k \geq 1$, a losing position has value 0. For a winning position, every lower-depth successor retains a positive heap of size at most H. By induction and the fact that $U_j(H)$ increases with j, its value is at most $U_{k-1}(H)$. The other moves contribute exactly the allowances in Table 2. Their sum gives $g(P) \leq U_k(H)$.

If $r(P) \geq 3$ is odd, there is no unique heap of minimum depth. The second allowance in that table is therefore absent, giving the refinement for $k \geq 1$. At depth zero, Lemma 2.1 gives

$$g(P) = 1 = U_0(H) - H,$$

which proves the refinement there as well. □

The bound from the smallest heap

Expanding the recurrence adds the allowances from each depth:

$$U_k(H) = H + \left\lfloor \frac{H}{2} \right\rfloor + \cdots + \left\lfloor \frac{H}{2^k} \right\rfloor + k + 1. \tag{11}$$

The first terms count the possible additional values at the successive depths. The term $k+1$ comes from the $+1$ at the base and at each recursive step.

Now choose $H = h_{\min}$, the smallest nonempty heap of the original position. Its depth is at least the minimum depth:

$$k = k(P) \le v_2(h_{\min}).$$

Thus 2^k divides $h_{\min}$, and every floor in (11) is an exact division. The contributions combine as

$$U_k(h_{\min}) = \underbrace{h_{\min} + \frac{h_{\min}}{2} + \cdots + \frac{h_{\min}}{2^k}}_{\text{allowances from successive depths}} + \underbrace{k+1}_{\text{the base and recursive } +1 \text{ terms}}.$$

Corollary 2.5 (The bound from the smallest heap)**.** *For a nonempty position with smallest heap $h_{\min}$ and minimum 2-adic depth k,*

$$\boxed{\begin{aligned} g(P) &\le U(h_{\min}, k) := 2h_{\min} - \frac{h_{\min}}{2^k} + k + 1 \\ &\le C(h_{\min}) := 2h_{\min} + \lfloor \log_2 h_{\min} \rfloor. \end{aligned}} \tag{12}$$

If $r(P) \ge 3$ is odd, the first bound improves by $h_{\min}/2^k$.

Proof. The geometric sum gives the first expression. The second inequality follows from

$$\frac{h_{\min}}{2^k} \ge 1, \qquad k \le \lfloor \log_2 h_{\min} \rfloor.$$

The refinement follows from (10). □

For example, when the smallest heap has size 4, the bound is built as follows:

Minimum 2-adic depth	Calculation	SG bound
0	$4+1$	5
1	$5+4/2+1$	8
2	$8+4/4+1$	10

Table 3: Building the bound when $h_{\min} = 4$.

At depth 2, if the number of heaps at minimum depth is odd and at least three, the allowance $4/4 = 1$ is absent and the bound improves from 10 to 9.

The ceiling $C(h_{\min})$ depends only on the smallest heap. Neither the number nor the sizes of the other heaps appear.

Proposition 2.6 (A bound for every descendant)**.** *Let P be a nonempty position and define*

$$h_{\max}(P) := \max_i h_i.$$

Every nonempty position R reached from P by any number of legal moves satisfies

$$\boxed{g(R) \le C\big(h_{\max}(P)\big) = 2h_{\max}(P) + \lfloor \log_2 h_{\max}(P) \rfloor\,.} \tag{13}$$

The empty position has SG value 0.

Proof. A legal move never increases a heap. Therefore every heap in a nonempty descendant R has size at most $h_{\max}(P)$, and in particular

$$h_{\min}(R) \le h_{\max}(P).$$

Corollary 2.5 gives

$$g(R) \le C\big(h_{\min}(R)\big) \le C\big(h_{\max}(P)\big),$$

because C is nondecreasing. The empty position has value 0. □

3 Bounding periods

The fixed board is $A = (h_1, \dots, h_{m-1})$, and n is the varying heap. The smallest fixed heap is $h_{\min}$; m includes the varying heap. We use the eventual period ω_A defined in Section 1: from some starting size onward, increasing n by ω_A preserves every SG value.

Our bound has the same structure as Morisawa's: the number of possible blocks of consecutive SG values multiplied by the comparison spacing, now denoted by W. We bound each component using the smallest fixed heap, apart from the SG periods of direct successor boards.

The ingredients of the period bound

As before, a move changes either the varying heap or a fixed heap. Write $A \xrightarrow{d} Q$ when removing d stones from one heap of A produces Q and d divides every other heap of A. On the full board (A, n), the removal must also divide n.

Successor value and meaning	Expression and reason
$g(A, n-d)$ Value after changing the varying heap	The removal must divide every fixed heap. Thus $d \mid D$, where $D = \gcd(h_1, \dots, h_{m-1})$. This successor uses an earlier SG value in the same sequence.
$g(Q, n)$ Value after changing a fixed heap	The move must satisfy $A \xrightarrow{d} Q$ and $d \mid n$. Its SG value comes from the smaller fixed board Q with the varying heap attached.

Combining the two types of successor values gives, for $n \ge D$,

$$g(A, n) = \operatorname{mex}\Big(\{g(A, n-d) : d \mid D\} \ \cup \ \{g(Q, n) : A \xrightarrow{d} Q,\ d \mid n\}\Big). \tag{14}$$

The largest removal from n is D, so a block of D consecutive SG values supplies every earlier value needed for the next calculation. Each entry has at most $C(h_{\min}) + 1$ possibilities. The comparison spacing W makes legal moves and the SG values of direct successor boards repeat together.

Quantity and meaning	Expression and reason
$0, \ldots, C(h_{\min})$ SG value range	Our SG bound gives $C(h_{\min}) = 2h_{\min} + \lfloor \log_2 h_{\min} \rfloor$. Including zero gives $C(h_{\min}) + 1$ possibilities for each entry in a block.
$J(h_{\min})$, J_A Increases in n that repeat legal moves	Every fixed-heap removal is at most $h_{\min}$, so $J(h_{\min}) = \operatorname{lcm}(1, \ldots, h_{\min})$ is a multiple of every such removal. With at least two fixed heaps, every removal also divides Morisawa's M_A. We therefore use the greatest common divisor $$J_A = \gcd\big(M_A, J(h_{\min})\big), \tag{15}$$ which is a common multiple of every legal fixed-heap removal amount. With one fixed heap, use $J_A = J(h_{\min})$.
D Number of consecutive SG values compared	The largest varying-heap removal is $D = \gcd(h_1, \ldots, h_{m-1}) \le h_{\min}$. The block $$\big(g(A, n - D + 1), \ldots, g(A, n)\big)$$ supplies all earlier values needed to compute $g(A, n + 1)$.

These three quantities are bounded using only $h_{\min}$. The SG periods of direct successor boards complete the comparison spacing:

Quantity and meaning	Expression and reason
$\omega_{Q_1}, \ldots, \omega_{Q_t}$ SG periods for direct successor boards	The list contains only the distinct nonempty boards Q reached by $A \xrightarrow{d} Q$. Each ω_Q is a chosen eventual period of $g(Q, n)$. No other smaller fixed boards supply values in the recurrence.
W Comparison spacing	The required common multiple is $$W = \operatorname{lcm}(J_A, \omega_{Q_1}, \ldots, \omega_{Q_t}).$$ It repeats legal moves and all direct-successor SG values together, once their repeating parts have begun.

If a move deletes the last fixed heap, its successor is the sole heap n, with SG value n. For $n > C(h_{\min})$, this value is too large to affect the parent's mex. Thus no period is needed for an empty fixed board. If there are no nonempty direct successors, take $W = J_A$.

Lemma 3.1 (Repeating legal moves)**.** *For $n \ge h_{\min}$, increasing n by J_A preserves every legal choice of heap and removal amount.*

Proof. A removal from a smallest fixed heap must fit in that heap. A removal from any other fixed heap must divide the unchanged smallest heap. In both cases $d \le h_{\min}$, so $d \mid J(h_{\min})$.

With at least two fixed heaps, a removal from h_i also divides the GCD of the other fixed heaps. It therefore divides M_A and hence J_A. Thus $d \mid n$ exactly when $d \mid n + J_A$. With one fixed heap, $d \mid J_A$ holds by definition. For the varying heap, the permitted removals are the divisors of $D \le h_{\min}$, which already fit. □

There are at most $[C(h_{\min}) + 1]^D$ different blocks. Comparing one more block than this number, at intervals of W, guarantees two matching blocks. Their separation is an eventual period, as the following proof shows.

Theorem 3.2 (An eventual-period bound)**.** *There is an eventual period satisfying*

$$\boxed{\omega_A \le \underbrace{W}_{\textit{comparison spacing}} \underbrace{[C(h_{\min}) + 1]^D}_{\textit{number of possible blocks}}.} \tag{16}$$

In particular,

$$\omega_A \le \underbrace{\operatorname{lcm}(\omega_{Q_1},\dots,\omega_{Q_t})}_{\text{SG periods of direct successors}} \underbrace{J(h_{\min})[C(h_{\min})+1]^{h_{\min}}}_{\text{depends only on } h_{\min}}. \tag{17}$$

The LCM of an empty list is taken to be 1.

Proof. Write $N = [C(h_{\min})+1]^D$. Choose n beyond the initial segments of all successor periods, with $n > \max(D, C(h_{\min}))$. Inspect the $N+1$ ordered blocks of D consecutive SG values ending at $n, n+W, \dots, n+NW$. Two blocks agree; their endpoints differ by qW for some $1 \le q \le N$.

At the next heap size, the fixed-heap moves supply the same values because their legality and successor values repeat after qW. The varying-heap moves refer to matching entries in the two blocks. Thus the mex gives the same next value. Repeating the argument proves that qW is an eventual period. Finally, $W \le J(h_{\min})\operatorname{lcm}(\omega_{Q_1},\dots,\omega_{Q_t})$ and $D \le h_{\min}$ give the separated bound. □

Comparison with Morisawa

Morisawa's counting argument uses a larger SG range and longer blocks. His comparison spacing ω' also includes periods of all nonempty fixed boards with fewer total stones, whereas W uses only direct successors. Assign the same chosen period to each board in both constructions. Then $W \mid \omega'$.

Quantity	Morisawa's proof	Present bound
SG value range	$0,\dots,mh_{\min}$	$0,\dots,C(h_{\min})$
Number of consecutive SG values compared	M_A+1	$D \le h_{\min}$
Number of possible blocks	$(mh_{\min}+1)^{M_A+1}$	$[C(h_{\min})+1]^D$
Increase in n that repeats legal moves	M_A	$J_A \mid J(h_{\min})$
SG periods included	All nonempty fixed boards with fewer total stones, using at most $m-1$ heaps	Only the nonempty direct successors $Q_1,\dots,Q_t$
Comparison spacing	ω'	W
Period upper bound	$\omega'(mh_{\min}+1)^{M_A+1}$	$W[C(h_{\min})+1]^D$

Table 4: Comparison with the counting argument in Morisawa's proof of Theorem 2.5 [1, Section 8]. Here $m \ge 3$ includes the varying heap.

Since $C(h_{\min}) < 3h_{\min} \le mh_{\min}$ and $D \le M_A$, both the SG value range and the number of consecutive SG values compared are reduced. Equation (17) shows exactly what remains: in that uniform bound, only the LCM of successor-board periods is not controlled by $h_{\min}$.

For a concrete comparison, take $A = (2,3)$. Its direct successors are $(1,3)$, $(2,2)$, and $(1,2)$, with periods $2,4,2$. Thus $h_{\min} = 2$, $D = 1$, $C(h_{\min}) = 5$, and

$$J_A = 2, \qquad W = \operatorname{lcm}(2,2,4,2) = 4.$$

Morisawa's larger list also includes the single fixed heap (4), since $4 < 2+3$. Its sequence has period 24; the other smaller boards have periods 2 or 4. Using these least periods gives $M_A = 6$ and $\omega' = 24$. The comparison is therefore

$$\begin{aligned} \text{Morisawa's counting bound} &= 24 \cdot 7^7 = 19{,}765{,}032, \\ \text{Present bound} &= 4 \cdot 6^1 = 24, \\ \text{Actual least period} &= 4. \end{aligned}$$

Keeping the same bounds throughout the successor chains

The smallest fixed heap may disappear. We therefore need a size limit that stays valid after any number of moves on the fixed board. The largest original fixed heap supplies that limit: no move can increase a heap.

Throughout this construction, $h_{\max}$ is the largest fixed heap in the original board A. It excludes the varying heap n and stays fixed throughout the induction. Every nonempty fixed descendant Q still contains a heap no larger than $h_{\max}$, even when n is arbitrarily large. Thus $h_{\min}(Q, n) \le h_{\max}$. Corollary 2.5, together with the monotonicity of C, therefore gives

$$\boxed{g(Q, n) \le C(h_{\max}) = 2h_{\max} + \lfloor \log_2 h_{\max} \rfloor.} \tag{18}$$

The current board can use its smaller bound $C(h_{\min})$. The bound $C(h_{\max})$ keeps one size limit valid for every nonempty fixed descendant.

The same replacement works for all three local ingredients of the period argument:

Quantity	Current fixed board A	Nonempty fixed descendant Q
SG value ceiling	$C(h_{\min})$	$C(h_{\max})$
Number of previous SG values needed	$D_A = \gcd(A) \le h_{\min}$	$D_Q = \gcd(Q) \le h_{\max}$
Increase in n that repeats legal moves	$J(h_{\min})$ suffices	$J(h_{\max})$ suffices

Table 5: One set of bounds remains valid even after the smallest fixed heap disappears. The varying heap n is not included in $h_{\max}$.

Each descendant has at least one positive fixed heap at most $h_{\max}$. This bounds its SG value, its largest varying-heap removal, and every fixed-heap removal. Once $n \ge h_{\max}$, all varying-heap removals fit, and adding $J(h_{\max})$ preserves the divisibility tests on n.

If a move removes the last fixed heap, the successor has SG value n. For $n > C(h_{\max})$, that value exceeds the parent's ceiling and cannot affect its mex. Thus only nonempty fixed successors need periods.

From a block bound to one common period

At each board, the earlier argument compares blocks of consecutive SG values. The table above bounds both the length of a block and the number of choices for each entry. It therefore bounds the number of different blocks using $h_{\max}$ alone.

Suppose T repeats all direct-successor SG sequences and legal moves. Two matching blocks, compared at intervals of T, give a parent period qT. The block count bounds q. To cover every possible q, we take the LCM of all integers up to that bound.

Quantity and meaning	Expression and reason
$N_{\max}$ Bound on the number of possible blocks	Each entry has at most $C(h_{\max})+1$ possible SG values, and at most $h_{\max}$ entries are needed. Thus $$N_{\max} = [C(h_{\max})+1]^{h_{\max}}.$$ Comparing $N_{\max}+1$ blocks gives two matching blocks separated by qT, where $1 \le q \le N_{\max}$.
$K(h_{\max})$ Common multiple of all possible multipliers	The multiplier q need not divide $N_{\max}$. Instead, use a number divisible by every possible q: $$K(h_{\max}) = \operatorname{lcm}(1,\dots,N_{\max}) = J(N_{\max}).$$ Every possible period qT then divides $TK(h_{\max})$.

Table 6: A bound on the block count gives a common multiple of all periods that one recursive layer may produce.

The distinction between a bound and a common multiple matters. For example, $3 \le 4$ does not mean that 3 divides 4. Although a parent period is at most $TN_{\max}$, that number need not itself be a period. The number $TK(h_{\max})$ is a period because it is a multiple of every possible qT.

Lemma 3.3 (One common period for the next layer)**.** *Suppose every heap of a nonempty fixed board is at most $h_{\max}$, and T is a positive multiple of $J(h_{\max})$. If T is an eventual period of every nonempty direct-successor sequence, then $TK(h_{\max})$ is an eventual period of the parent sequence.*

Proof. Use T as the comparison spacing in Theorem 3.2. The actual block count is at most $N_{\max}$, so the proof gives a period qT with $1 \le q \le N_{\max}$. Since $q \mid K(h_{\max})$, $TK(h_{\max})$ is a multiple of this period and is a period too. This argument applies after all successor sequences have reached their repeating parts. □

Within one layer, the number of successor boards adds no further factor. If they all have a common period T, their values all repeat after T, whether there are three successors or three thousand. The next factor comes from continuing the parent's own recurrence, rather than from counting these successors.

Why the recursive bound contains a power

LCM combines the successor periods at one layer. Moving up to a parent can introduce a new multiplier *outside* that LCM. This is why a bound on one layer does not yet bound every layer by the same number.

First consider a chain in which each layer uses the same multiplier N. This illustrates the accumulation; the actual multipliers may be different. Write J for the increase that repeats legality. If the starting period is $w_0 = JN$, then

$$w_1 = \operatorname{lcm}(J, JN)\, N = JN^2,$$
$$w_2 = \operatorname{lcm}(J, JN^2)\, N = JN^3.$$

The LCM keeps the previous power of N, because JN^2 already contains J. The new multiplier then adds another factor of N.

For example, if $J = 6$ and $N = 4$, the calculation is:

Layer	Calculation	Illustrative value
Starting layer	$6 \cdot 4$	24
Next layer	$\operatorname{lcm}(6,24) \cdot 4$	96
Following layer	$\operatorname{lcm}(6,96) \cdot 4$	384

Table 7: Illustrative multiplier accumulation. These numbers explain the recurrence; they are not computed Divisor Nim periods.

Repeated entries in $\operatorname{lcm}(J, N, N, N)$ add nothing. In the constant-multiplier illustration above, each new layer instead multiplies by another factor of N, giving JN, JN^2, JN^3. These equalities are exact for that illustration, and every term has the same prime divisors as JN.

The actual block argument does not establish that every multiplier equals $N_{\max}$. If T repeats legal moves and all direct-successor SG sequences, the argument produces a parent period qT, where

$$1 \le q \le N_{\max}, \qquad qT \le TN_{\max}.$$

This inequality bounds the period's size. It does not imply that qT divides $TN_{\max}$, or that its prime divisors divide $TN_{\max}$.

To obtain a guaranteed common period, we use

$$K(h_{\max}) = \operatorname{lcm}(1, \ldots, N_{\max}).$$

Since every possible q divides $K(h_{\max})$, the period qT divides $TK(h_{\max})$. Thus $TK(h_{\max})$ is an eventual period regardless of which multiplier occurs. This divisibility is what supports the common-period induction and the subsequent prime-factor conclusion.

We now build that guaranteed period layer by layer. The fixed board (1) starts the induction. Each move on a fixed heap removes at least one stone, so the total fixed stones minus one bounds the number of layers before only one fixed stone remains.

Quantity and meaning	Expression and reason
$T_0(h_{\max})$ Starting common period	The sequence $g(1, n)$ has period 2. Combine it with the increase that repeats legal moves: $$T_0(h_{\max}) = \operatorname{lcm}(2, J(h_{\max})).$$ This is a period for $(1, n)$ and repeats legality for every fixed board under the same size limit.
$s(A)$ Number of recursive layers allowed	For a nonempty fixed board, $$s(A) = \sum_{h \in A} h - 1.$$ Repeated heaps are counted separately. A nonempty fixed successor Q satisfies $s(Q) \le s(A) - 1$. The case $s = 0$ is exactly the board (1). This counts fixed stones, excluding n; it is not the number of common halvings.

Table 8: The base of the induction and a bound on its length.

For a fixed value of $h_{\max}$, the common periods build as follows:

Layer bound	Why it covers the next board	Guaranteed common period
0	The fixed board is (1).	$T_0(h_{\max})$
1	All nonempty successors are covered at layer 0; one more layer contributes a factor $K(h_{\max})$.	$T_0(h_{\max})K(h_{\max})$
2	All nonempty successors are covered by the preceding common period; apply the same one-layer argument.	$T_0(h_{\max})K(h_{\max})^2$
s	Every successor has a smaller layer bound, so s factors cover the whole induction.	$T_0(h_{\max})K(h_{\max})^s$

Table 9: The exponent counts repeated applications of the one-layer period argument, not the number of successors at a layer.

Combining the starting period with one common multiplier per allowed layer gives the explicit period.

Proposition 3.4 (An explicit period without successor periods)**.** *Fix a positive heap-size limit $h_{\max}$. For every nonempty fixed board A whose heaps are at most this limit,*

$$\boxed{\underbrace{T_0(h_{\max})}_{\text{starting common period}} \cdot \underbrace{K(h_{\max})^{s(A)}}_{\text{one factor per allowed layer}}} \tag{19}$$

is an eventual period of $g(A, n)$. Its least eventual period satisfies

$$\boxed{\omega_A \mid T_0(h_{\max})K(h_{\max})^{s(A)}.} \tag{20}$$

Proof. Induct on $s = s(A)$, keeping the original size limit $h_{\max}$ fixed. If $s = 0$, then $A = (1)$ and $g(1, n)$ is 1 for even n and 0 for odd n. Thus $T_0(h_{\max})$ is a period.

For $s \geq 1$, each nonempty direct successor Q has $s(Q) \leq s-1$. By induction it has period $T_0(h_{\max})K(h_{\max})^{s(Q)}$. All these periods divide

$$T = T_0(h_{\max})K(h_{\max})^{s-1}.$$

Therefore T repeats every nonempty successor sequence and legal move. The one-layer lemma gives the parent period

$$TK(h_{\max}) = T_0(h_{\max})K(h_{\max})^{s}.$$

Finally, the least eventual period divides every eventual period: on the repeating tail, any shift preserving the sequence must be a multiple of its shortest cycle. □

For the original board, choose its actual largest fixed heap as $h_{\max}$. The formula now uses only that heap and the total fixed stones. It requires no separate calculation of successor periods. Equation (16) remains sharper when those periods are known.

What this proves about prime factors

The number of layers can increase the *powers* of prime factors in this guaranteed period, but it cannot add primes outside a fixed collection determined by $h_{\max}$.

Every prime dividing $K(h_{\max}) = \operatorname{lcm}(1, \ldots, N_{\max})$ is at most $N_{\max}$. Also $N_{\max} \geq \max(2, h_{\max})$, so $T_0(h_{\max})$ divides $K(h_{\max})$ and introduces no other primes. Equation (20) therefore gives:

Corollary 3.5 (A fixed set of possible prime divisors)**.** *For fixed $h_{\max}$, every prime divisor of the least eventual period of any board with fixed heaps at most $h_{\max}$ is at most $N_{\max} = [C(h_{\max}) + 1]^{h_{\max}}$, independently of heap count.*

Adding heaps no larger than $h_{\max}$ keeps this same collection of possible primes. It can change which primes in the collection actually occur. The proof does not show that a prime disappearing from one period can never return, or that its exponent cannot grow. For example, powers $2, 4, 8, 16$ all have the same prime divisor but grow without bound if the exponents keep increasing.

Long successor chains also need not produce long periods. Any number of fixed heaps of size 1 gives period 2, although $s(A)$ grows with their number. The exponent in our bound allows growth at every layer; it does not say the actual period grows.

One possible improvement is to retain only the successor values that affect the mex. Values above the parent's SG ceiling can be ignored. A successor sequence can also be omitted if each relevant value it supplies is already supplied by a retained legal move at that same n. This condition must hold throughout the repeating tail. Bounding how many layers remain after these reductions is one route toward a period independent of heap count.

A conjecture and computed periods

The explicit bound leaves prime exponents dependent on the number of layers. The computations suggest a much smaller common period: twice the LCM of all positive integers up to the largest fixed heap.

Conjecture. For every nonempty fixed board A, its least eventual SG period satisfies

$$\boxed{\omega_A \mid 2J(h_{\max}(A)) = 2\,\mathrm{lcm}(1,\ldots,h_{\max}(A)).} \tag{21}$$

Equivalently, $2J(h_{\max}(A))$ is an eventual period of $g(A,n)$. The largest fixed heap excludes n. The starting size at which the repetition begins may still depend on the whole fixed board.

This is stronger than asking for some bound $F(h_{\max})$: it gives a particular common period. It also specifies the possible prime powers. For an odd prime p, the LCM contains the largest power $p^a \le h_{\max}$. Multiplication by 2 allows one extra factor of 2.

Quantity	Proved here	Proposed in the conjecture
Common eventual period	$T_0(h_{\max})K(h_{\max})^{s(A)}$	$2J(h_{\max})$
Possible prime divisors	At most $N_{\max}$, independent of heap count	At most $h_{\max}$, apart from 2 when $h_{\max}=1$
Powers of those primes	The bound can grow with $s(A)$	Bounded using $h_{\max}$ alone

Table 10: The conjecture would remove the remaining recursive-layer dependence, not merely restrict the set of possible primes.

As a baseline, the least eventual periods with one fixed heap are listed below. Morisawa records the values for $h=1,\ldots,6$ from earlier work and proves the value for $h=7$ [2, Theorem 3 and Corollary 1]. Our recurrence calculation reproduces these entries and gives the displayed values through $h=15$. These are least periods, not upper bounds.

Fixed heap h	Least eventual period of $g(h,n)$
1	2
2	4
3	2
4–6	24
7	120
8–10	1,680
11	5,040
12–14	55,440
15	720,720

Table 11: Least eventual periods with one fixed heap. Ranges group consecutive heap sizes having the same period.

Repetition was certified by checking a common shift that preserves all divisibility tests and matching blocks for every nonempty fixed descendant. Induction on fixed stones and then on n extends the matching blocks indefinitely. For $h \le 15$, a common shift of 720,720 passes these checks from $n = 10{,}000$. Testing the divisors of that shift within a full cycle gives each least period above.

We also checked all fixed boards in the following finite ranges. Heap order does not matter, but repeated sizes are included.

Allowed fixed heap sizes	Maximum fixed heap count	Boards checked
$1,\ldots,8$	12	125,969
$1,\ldots,12$	6	18,563
$1,\ldots,14$	4	3,059
$1,\ldots,16$	3	968

Table 12: Certified period checks supporting (21). The ranges overlap.

After removing overlaps, these tests contain 143,059 distinct nonempty fixed boards. Every computed least period divides $2J(h_{\max}(A))$, using that board's own largest fixed heap. The checks certify eventual repetition as above; they do not prove the conjecture for arbitrarily many heaps.

The single-fixed-heap period need not itself be a universal period for all boards under that heap limit. For example,

$$g(2,3,n) = \begin{cases} 0, & n \text{ odd}, \\ 2, & n \equiv 0 \pmod 4, \\ 3, & n \equiv 2 \pmod 4. \end{cases}$$

Its period is 4, whereas $g(3,n)$ has period 2. Both divide the conjectured common period $2J(3) = 12$.

Adding heaps need not change the least period in one direction. The following certified example first shortens the period and then partly reverses that reduction:

Fixed board A	Least eventual period of $g(A,n)$
(8)	1,680
$(1,8,8,8)$	2
$(1,4,8,8,8)$	12

Table 13: Adding fixed heaps can shorten a period and then increase it.

In the last step, the smallest heap stays 1, the largest stays 8, the GCD stays 1, and the LCM stays 8. Every divisor of the added heap 4 already divides 8, yet the period rises from 2 to 12. For these rows, the common shift 1,680 was checked from $n = 216$ on all fixed descendants; full-cycle checks then give the least periods. Thus unchanged heap extremes and divisors do not guarantee an unchanged period. The conjecture asks instead for a common period that survives arbitrary additions of heaps of size at most $h_{\max}$.

4 Composition: exact results, then bounds

Join two boards by putting all their heaps under one shared divisor rule, and write the result as $A \sqcup B$. A move on a heap from A must now satisfy the restrictions from B too. This is different from playing two independent games side by side: the join changes the legal moves themselves. Consequently, $g(A \sqcup B)$ need not equal $g(A) \oplus g(B)$, where $\oplus$ is the bitwise exclusive-or used for independent games.

Exact Grundy values when the join has at least two odd heaps

The numbers o_A and o_B count the odd heaps in A and B before any halving. Joining the boards adds these counts. If the total is at least two, its parity gives the exact SG value.

Proposition 4.1 (Exact SG composition in the odd-heap case)**.** *If $o_A + o_B \geq 2$, then*

$$\boxed{g(A \sqcup B) = \begin{cases} 0, & o_A + o_B \text{ is even}, \\ 1, & o_A + o_B \text{ is odd}. \end{cases}}$$

Proof. Apply Lemma 2.1 to the joined board. □

One odd heap from each board is enough. The sizes of those heaps and the sizes and number of even heaps do not change this value.

This also gives a replacement rule. A board A with at least two odd heaps can be replaced by $(1,1)$ if its odd count is even, or by $(1,1,1)$ if its odd count is odd. For every future board B, the original and replacement joins retain at least two odd heaps and have the same odd-count parity. Their SG values therefore agree.

Exact winning and losing outcomes for every join

For each nonempty board, retain its minimum 2-adic depth and the number of heaps at that depth. These are (k_A, r_A) for A and (k_B, r_B) for B. If one board reaches an odd heap first under common halving, it determines where halving stops for the join. If both reach odd heaps together, their counts add.

Quantity and meaning	Expression and reason
k Minimum 2-adic depth of the join	$k = \min(k_A, k_B)$. Common halving stops as soon as either board has an odd heap.
r Number of joined heaps at depth k	Only heaps at the smaller depth are counted; if the depths agree, both boards contribute: $$r = \begin{cases} r_A, & k_A < k_B, \\ r_B, & k_B < k_A, \\ r_A + r_B, & k_A = k_B. \end{cases}$$

Combining these quantities with Morisawa's outcome criterion gives

$$\boxed{g(A \sqcup B) = 0 \quad \Longleftrightarrow \quad r \text{ is even.}} \tag{22}$$

An odd count means the join is winning.

For any number of boards, take the smallest depth and add the counts from boards at that depth. This does not depend on the order or grouping of the joins, so the composition rule is commutative and associative. Empty boards can be omitted.

Only the parity of r is needed for outcomes. Two boards with the same minimum depth and the same parity of the count at that depth can replace one another in every future join without changing its outcome. A board joined to either one has smaller, equal, or larger depth; in each case, the count rule gives the same final parity.

Thus full SG composition is exact in the odd-heap case, and outcome composition is exact for all boards. For the remaining positive SG values, we use the bounds already proved.

The SG bounds compose too

The SG bound uses two inputs: the smallest heap $h_{\min}$ and the minimum 2-adic depth k. Joining boards takes the minimum of each input. We already have k from the outcome calculation; adding the smallest heap gives the bound.

Quantity and meaning	Expression and reason
$h_{\min}$ Smallest heap of the join	$h_{\min} = \min\big(h_{\min}(A), h_{\min}(B)\big)$. The smallest heap in the combined board is the smaller of the two original smallest heaps.
$U(h_{\min}, k)$ SG ceiling for the join	Substituting the joined smallest heap and minimum depth into the bound from Section 2 gives $$U(h_{\min}, k) = 2h_{\min} - \frac{h_{\min}}{2^k} + k + 1.$$ The count r then determines whether the value is exactly zero or the ceiling can be tightened.

Table 14: The quantities used to bound the SG value of a join.

The ceiling $U(h_{\min}, k)$ increases with $h_{\min}$, since its coefficient $2 - 2^{-k}$ is positive. It also increases with k, because

$$U(h_{\min}, k+1) - U(h_{\min}, k) = h_{\min}/2^{k+1} + 1 > 0.$$

Taking the minimum of each input therefore cannot increase the ceiling.

Proposition 4.2 (A bound preserved by joining)**.** *For nonempty boards A, B, let $(h_{\min}, k, r)$ be their joined summary. Then*

$$\boxed{g(A \sqcup B) \le U(h_{\min}, k) \le \min\{U(h_{\min}(A), k_A), U(h_{\min}(B), k_B)\}.} \tag{23}$$

If r is even, the exact value is 0. If $r \ge 3$ is odd, the first ceiling improves to $U(h_{\min}, k) - h_{\min}/2^k$.

Proof. Apply Corollary 2.5 and (22) to the joined board. The comparison with the two original ceilings follows from the increase of U in each input. □

The smallest heap and smallest depth may come from different boards, so their combination can improve on both original ceilings. For example, let $A = (4, 16)$ and $B = (6, 16)$. Their ceilings are 10 and 11. The join takes $h_{\min} = 4$ from A and $k = 1$ from B, giving 8. Adding heaps of size 2 then illustrates the role of the count.

Board	$h_{\min}$	k	r	SG bound or exact value
$A = (4, 16)$	4	2	1	10
$B = (6, 16)$	6	1	1	11
$A \sqcup B$	4	1	1	8
$A \sqcup B \sqcup (2, 2)$	2	1	3	$5 - 1 = 4$
$A \sqcup B \sqcup (2, 2, 2)$	2	1	4	0 (exact)

Table 15: The positive entries are upper bounds. The last entry is the exact losing value.

In the row with $r = 3$, Corollary 2.5 improves $U(2, 1) = 5$ by $h_{\min}/2^k = 2/2 = 1$, giving the bound 4.

Finally, start with any fixed nonempty board A and keep adding boards. Repeated application of (23) gives

$$\boxed{g(A \sqcup B_1 \sqcup \cdots \sqcup B_t) \le U(h_{\min}(A), k_A) \quad \text{for every } t \ge 0.} \tag{24}$$

The actual value may rise or fall as heaps are added. The ceiling provided by the original board remains valid, regardless of how many additional heaps appear or how large they are.

Composing period bounds

For periodicity, A and B are fixed boards and n is one varying heap attached to their join. The explicit common period uses the largest fixed heap and the total fixed stones. Joining boards takes the larger heap size and adds the stone counts.

Quantity and meaning	Expression and reason
$h_{\max}$ Largest fixed heap of the join	The largest heap comes from one of the two boards: $$h_{\max} = \max(h_{\max}(A), h_{\max}(B)).$$ This same size limit remains valid in every fixed descendant of the join.
$s(A \sqcup B)$ Joined number of allowed layers	Each s is the total fixed stones minus one. Hence $$s(A \sqcup B) = s(A) + s(B) + 1.$$ The extra 1 corrects for subtracting one in each input count.

Substituting these joined quantities gives a common period without calculating any successor periods.

Corollary 4.3 (A period bound for joined boards)**.** *For nonempty fixed boards A, B, use their joined largest heap $h_{\max}$. Then*

$$\boxed{\omega_{A\sqcup B} \mid T_0(h_{\max})K(h_{\max})^{s(A)+s(B)+1}.} \tag{25}$$

The expression on the right is a guaranteed eventual period.

Proof. Apply Proposition 3.4 to $A \sqcup B$ and use the two quantities in the table. □

For several boards, take the largest fixed heap and add all fixed stones. These operations do not depend on the grouping or order of the joins. If heaps no larger than $h_{\max}$ are added, the possible prime collection remains unchanged. The guaranteed period can still grow through its exponent s.

Under Conjecture (21), the join would instead have common period $2J(h_{\max})$, independent of those stone counts. This would compose a period guarantee from the largest fixed heap alone. It would not imply that the actual least period always increases or always decreases when boards are joined.

5 Conclusions and open questions

The smallest nonempty heap bounds the current SG value by $C(h_{\min}) = 2h_{\min} + \lfloor \log_2 h_{\min} \rfloor$, independently of all other heap sizes and counts. The largest initial heap supplies a bound for every descendant. Joining boards preserves and can tighten the SG ceiling; minimum 2-adic depth and its count parity determine every joined outcome.

For periods, the smallest fixed heap controls the local terms. The largest original fixed heap keeps one set of bounds valid throughout the successor chains. The resulting common period is

$$T_0(h_{\max})K(h_{\max})^{s(A)}, \qquad s(A) = \sum_{h\in A} h - 1.$$

LCM combines all successors at one layer without a factor for their number. Moving up a layer may add another multiplier, which is why the exponent still depends on the total fixed stones. The possible prime divisors nevertheless belong to one finite set determined by $h_{\max}$ alone.

The two principal conjectures are:

Conjecture and meaning	Precise claim and significance
Sharper SG bound	For every nonempty position P, $\boxed{g(P) \le 2h_{\min}(P).}$ This would remove the remaining logarithmic term.
Common period independent of heap count	For every nonempty fixed board A, $\boxed{\omega_A \mid 2J(h_{\max}(A)).}$ Here $J(h_{\max}) = \operatorname{lcm}(1, \ldots, h_{\max})$ and $h_{\max}(A)$ excludes the varying heap n. This would bound both the primes and their powers using the largest fixed heap alone.

The period conjecture holds for every board in the certified finite ranges reported above. A proof must still explain why arbitrarily many fixed heaps cannot keep adding new powers through recursive layers. Removing successor values that do not affect the mex, and understanding repeated copies of a heap, may help control those layers. The size at which eventual repetition begins may still depend on the whole fixed board.

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